A

IGHD

Count
0 5 10 15

D4-17 D4-11* D4-23 D5-24 D5-18* D5-12 D7-27 D1-14 D1-1 D1-20 D1-7 D1-26 D2-2 D2-15 D2-21 D2-8 D3-16 D3-22 D3-10 D3-3 D3-9 D6-19 D6-13 D6-25 D6-6

Subgroup

5' RSS

Usage
0.2
0.15
0.1
0.05
0

3' RSS

CAC?GTG A??C?G?C???T ?CA?AA?CC

CACAGTG A??????C???? TC??AAAC?

B

IGHJ

Usage
0.5 0.4 0.3 0.2 0.1 0

....T.. TGAC.ACAATGCCAGGACCC.A GGC..G.A.T
....T.. TGAC.ATAATGCCAGGACCC.A GGC..G.A.T
....GCC TCTGCCCT.CTGCTTCTCCCAT ACA....CA.
....GCC TCTGCCTT.CTGCTTCTCCCAT ACA....CA.
....T.. TGGG.GG..CCATTAAGGGGTG CAC....A..
....GG. ACAC.GT..CTGTT.CTGCC.A GAC.C..A..
....GG. ACAC.GT..CTGTT.CTGCC.A GAC.T..A..
....A.. GCAG.ATGTCCATC.TCACC.C ACA....C..
...GG.. AGTC.GC..TGAGC.AGGGG.T ACAG...C..
..TGG.. AGTC.GC..TGAGC.AGGGG.T ACAG...C..

CACA?TG ????A??CC?????C?????C? ???AAAA?CC

Subgroup
J2 J5 J1 J4 J3 J6

C

IGKJ

Usage
0.3 0.25 0.2 0.15 0.1

....... TTAAC.TAATTACTTTCCC..T A.......TC
......A TTAAC.TACTTTC-CCCTTA-- -.......TC
....A.. GTTCC.CTTAACTTCCCTC..A T.......CT
....... ATTCG.CTTAACTTTTCCCT.T ACA....CCT
....... AGAGC.CTCCATTGTCTTG..G A.......CC
....... AGAGC.CTCCATTGTCTTG..G T.......CC
....... AGGGA.CTCACCCTTTCCC..C A.......CC
....... AGGGA.CTTACCCTTT-CC..C A.......CC

CACAGTG ?????T??????????????CT? ?ACAAAAA??

Subgroup
J2 J3 J5 J1 J4

D

IGLJ

Usage
0.8 0.6 0.4 0.2 0

...T... .T.T.AC.CTGC ...C.....
...T..A .T.T.AC.CTGC ...C.....
...A... ..TG..GTT..G ..CA.....
...A... ..TG..G.T..G ..CA.....
...A... C.C..C ...C.....
...A... T.C..C ...C.....
...T... G.T..T ...A..G..
...T... G.T..T ...A.....
...C... G.T..T ...A.....

CAC?GTG ACACAG?C?CA? ACA?AAACC

Subgroup
J1 J2 J3-3 J3-1 J3-2 J3-4 J6 J7